

Correction — Exercice 16 : Navbar + composants Bootstrap

Solution ● — navbar enrichie + modale + breadcrumb +

alert

```
<!DOCTYPE html>
<html lang="fr">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Navbar + composants</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-primary">
  <div class="container">
    <a class="navbar-brand" href="#">MonSite</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>

    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav me-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Accueil</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Produits</a></li>

        <li class="nav-item dropdown">
          <a class="nav-link dropdown-toggle" href="#" role="button" data-bs-toggle="dropdown">Compte</a>
          <ul class="dropdown-menu">
            <li><a class="dropdown-item" href="#">Profil</a></li>
            <li><a class="dropdown-item" href="#">Paramètres</a></li>
            <li><hr class="dropdown-divider"></li>
            <li><a class="dropdown-item" href="#">Déconnexion</a></li>
          </ul>
        </li>
      </ul>

      <form class="d-flex" role="search">
        <input class="form-control me-2" type="search" placeholder="Rechercher">
        <button class="btn btn-outline-light" type="submit">OK</button>
      </form>
    </div>
  </div>
</nav>

<main class="container py-4">

  <nav aria-label="breadcrumb">
    <ol class="breadcrumb">
      <li class="breadcrumb-item"><a href="#">Accueil</a></li>
```

```

    <li class="breadcrumb-item"><a href="#">Produits</a></li>
    <li class="breadcrumb-item active" aria-current="page">Détail</li>
  </ol>
</nav>

<h1>Détail du produit</h1>

<div class="alert alert-success alert-dismissible fade show" role="alert">
  Produit enregistré avec succès.
  <button type="button" class="btn-close" data-bs-dismiss="alert" aria-label="Fermer"></button>
>
</div>

<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#monModal">
  Ouvrir la modale
</button>

<div class="modal fade" id="monModal" tabindex="-1" aria-labelledby="monModalLabel" aria-hidden="true">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title" id="monModalLabel">Nous contacter</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <form>
        <div class="modal-body">
          <div class="mb-3">
            <label for="mail" class="form-label">Email</label>
            <input type="email" id="mail" class="form-control" required>
          </div>
          <div class="mb-3">
            <label for="msg" class="form-label">Message</label>
            <textarea id="msg" class="form-control" rows="3"></textarea>
          </div>
        </div>
        <div class="modal-footer">
          <button type="button" class="btn btn-secondary" data-bs-dismiss="modal">Annuler</button>
          <button type="submit" class="btn btn-primary">Envoyer</button>
        </div>
      </form>
    </div>
  </div>
</div>
</main>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Solution ● — app.html (offcanvas + toast + accordéon + carousel + tooltips)

Extraits clés (HTML complet long, on se concentre sur les parties spécifiques).

Offcanvas déclenché par la navbar

```
<button class="btn btn-outline-light" data-bs-toggle="offcanvas" data-bs-target="#menuAdmin">
  Menu admin
</button>

<div class="offcanvas offcanvas-start" id="menuAdmin" tabindex="-1">
  <div class="offcanvas-header">
    <h5 class="offcanvas-title">Administration</h5>
    <button class="btn-close" data-bs-dismiss="offcanvas"></button>
  </div>
  <div class="offcanvas-body">
    <ul class="list-group">
      <li class="list-group-item"><a href="#">📊 Tableau de bord</a></li>
      <li class="list-group-item"><a href="#">👤 Utilisateurs</a></li>
      <li class="list-group-item"><a href="#">⚙️ Paramètres</a></li>
      <li class="list-group-item"><a href="#">📁 Fichiers</a></li>
    </ul>
  </div>
</div>
```

Toast déclenché par JS

```
<div class="toast-container position-fixed bottom-0 end-0 p-3">
  <div id="monToast" class="toast" role="alert">
    <div class="toast-header">
      <strong class="me-auto">Notification</strong>
      <button class="btn-close" data-bs-dismiss="toast"></button>
    </div>
    <div class="toast-body">Action effectuée avec succès.</div>
  </div>
</div>

<button class="btn btn-primary" id="btnNotif">Notifier</button>

<script>
document.getElementById('btnNotif').addEventListener('click', () => {
  const toast = bootstrap.Toast.getOrCreateInstance(document.getElementById('monToast'));
  toast.show();
});
</script>
```

Accordéon always-open (pas de `data-bs-parent`)

```
<div class="accordion" id="faq">
  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button" data-bs-toggle="collapse" data-bs-target="#q1">
        Quelle est votre politique de retour ?
      </button>
    </h2>
    <div id="q1" class="accordion-collapse collapse show">
      <div class="accordion-body">30 jours sans condition.</div>
    </div>
  </div>
  <!-- Pas de data-bs-parent = always-open -->
  <!-- Répéter pour q2, q3, q4 -->
</div>
```

Tooltips (initialisation JS requise)

```
<button class="btn btn-secondary" data-bs-toggle="tooltip" title="Sauvegarder maintenant">💾 </button>

<script>
// Bootstrap n'active PAS les tooltips automatiquement, il faut les initialiser
const tooltipEls = document.querySelectorAll('[data-bs-toggle="tooltip"]');
tooltipEls.forEach(el => new bootstrap.Tooltip(el));
</script>
```

Points clés

♦ Les composants Bootstrap et leur dépendance JS

Composant	JS requis	Popper	Initialisation auto
Modal	✓	—	via <code>data-bs-toggle</code>
Dropdown	✓	✓	via <code>data-bs-toggle</code>
Collapse / Navbar	✓	—	via <code>data-bs-toggle</code>
Offcanvas	✓	—	via <code>data-bs-toggle</code>
Carousel	✓	—	via <code>data-bs-ride</code>
Tab	✓	—	via <code>data-bs-toggle</code>
Toast	✓	—	JS manuel (<code>.show()</code>)
Tooltip / Popover	✓	✓	JS manuel (<code>new bootstrap.Tooltip(el)</code>)
Alert dismiss	✓	—	via <code>data-bs-dismiss</code>

→ Toast, Tooltip et Popover sont **opt-in** pour ne pas alourdir les pages qui n'en ont pas besoin.

⚠ Ne pas oublier le JS bundle

Le HTML seul avec les `data-bs-*` **ne fonctionne pas** — il faut inclure `bootstrap.bundle.min.js` (qui contient Popper), sinon les interactions restent inertes et silencieuses (pas d'erreur console visible tant que l'utilisateur ne clique pas).